

Self-practice 1 - Multivariate linear regression

- Implement multivariate linear regression in Python to predict house price based on size and number of bedrooms. Based on collected data, build the algorithm to learn a model for predicting house price.
- Each sample consists of area, number of bedrooms and price.
- Complete the #TODO list in the attached notebook, "Data\LinearRegression.ipynb". The notebook's structure consists of 3 main parts:
- Part 1: Read and Analyze data:
 - Read and do basic analyze on the data.
 - Students need to make comments on the data based on tables and graphs.
- Part 2: Helper Functions:
 - Build necessary functions and class to prepare for model training and predict.
 - Students need to complete the missing code in following functions:
 - `standardize_train_set` function: standardize the features of train set based on their mean and standard deviation.
 - `standardize_test_set` function: standardize the features of test set based on their mean and standard deviation in train set.
 - `LinearRegression` class: used to build a linear regression model.
 - `calculate_absolute_error` function: calculate absolute error of model on training set in training progress.
 - `gradient` function: calculate gradient
 - `gradient_descent` function: calculate gradient descent
 - `predict` function: predict for new input
- Part 3: Main Function:
 - Training model on training set, predict new value with new input, save model and result in a file.
 - Students need to complete the missing code and make comments on the model based on the graph.
- Data: training data set and config file are attached at "Data\training_data.csv" and "Data\config.json".